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BRIEF REPORT

Maternal Behavior and the Milk Ejection Reflex in a Precocial Murid *(Acomys cahirinus)*

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Four litters of the precocial murid *Acomys cahirinus* were observed for 20 days in a study of mother–infant interactions. The nursing posture was similar in form to that seen in other rodents, and the pups showed responses of the kind that rats and mice characteristically show in response to milk ejection. Immediately after parturition nursing occupied most of the time during which observations were made, but the time spent nursing declined rapidly from day to day. The sequence of milk ejection inferred from the response of the pups was also similar to that of rats and mice, with a mean interval between milk ejections of 5 min 15 sec.

We report here some observations concerning mother–infant interaction in the spiny mouse, *Acomys cahirinus*. This is a species of a genus unique among murid rodents in that its young are born precocial, with sensory and motor capacities well developed at birth. There is no information available on its nursing behavior.

The animals were obtained from Parslow's Hamster Farm, Great Bookham, Leatherhead, Surrey, England, when they were 60 days old. They were housed in a large acoustically isolated chamber separated by one-way glass from an observation chamber. Illumination was by dim red light from 1000 until 2200 hr and by white light for the rest of the day.

Females were kept paired with males until the birth of a litter. The cages were checked daily and the day on which the litter was first found counted as Day 1. Data are presented from four litters; one with one, one with two, and two with three pups. The animals were easily disturbed so three litters were not touched at all; the fourth litter was weighed every few days and the growth of the pups was linear from 7.2 to 26.2 g per pup over the first 21 days. The eyes were open and the ears unfolded on the day of birth, and the pups ate dry food from Day 2 or 3 of life.

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The litters were observed at least 6 days a week for the first 20 days, for 105 min each day. One hour's observation took place in the light phase (900 to 1000 hr) and the rest in the dark phase (1130 to 1215 hr). These periods were chosen as a result of a pilot study which showed them to be peak periods of mother–offspring interaction. Each cage was observed every 80 sec. The mother's behavior at the time of observation was classified as follows: nursing (at least one pup attached to a nipple); licking (the pups); grooming (herself); eating/drinking; general activity; lying still/sleeping. Results were analyzed into blocks of 4 days. In addition to these daily observations, the litter with three pups was also observed for longer periods during the afternoon. During these afternoon sessions the sequence of nursing behavior was established, and the intervals between milk ejections were measured. Accurate timings were made on the afternoons of Days 3 and 7 and form the basis for Fig. 2.

Although the animals were initially given nesting material of straw, cotton waste, and paper, no nests were built; this accords with other reports (Porter & Ruttle, 1975). Licking of the pups was generally seen before or after nursing bouts, and occupied 6.2% of the time. Other than that spent nursing, the rest of the time was taken up in grooming (9.25%), general activity (16.3%), eating (9.3%), and lying still/sleeping (17.7%); these are percentages averaged over the first 20 days for all four litters.

Nursing itself occupied 41.5% of the time overall, but the proportion of time spent nursing was actually much higher at first and decreased from day to day (see Fig. 1). Figure 2 shows the sequence of nursing, and of milk ejection, for the afternoons of Days 3 and 7. These data were established in the following way. There is in the rat and mouse a characteristic nursing posture, which the mother adopts in response to one or more of the pups attaching to the nipple. The mother stands absolutely still over the pups, with her back arched so that her weight does not rest on the litter. We saw an identical nursing posture in the spiny mouse, and "nursing" in Fig. 1 refers to the time spent holding the nursing posture. In the rat, the pups lie very still on the nipple except during the milk ejection, which elicits synchronously from all the pups a characteristic sequence of responses, including pushing at the nipple, stretching and then moving from one nipple to another (Drewett, Wakerley, & Statham, 1974). The spiny mouse pups showed exactly the same pattern of behavior, and since the pups were large and their behavior was not hidden by a nest it was very easy to record the occurrence of each milk ejection reliably by observing the behavior of the pups; the sequences of milk ejections shown in Fig. 2 was determined in this way. This procedure makes the assumption that the correlations between the pups' behavior and the milk ejection are the same as those found in rats (and mice: Vorherr, Kleeman, & Lehman, 1967). The assumption could be checked directly by oxytocin infusions in anesthetized mothers, as in Drewett et al. (1974).

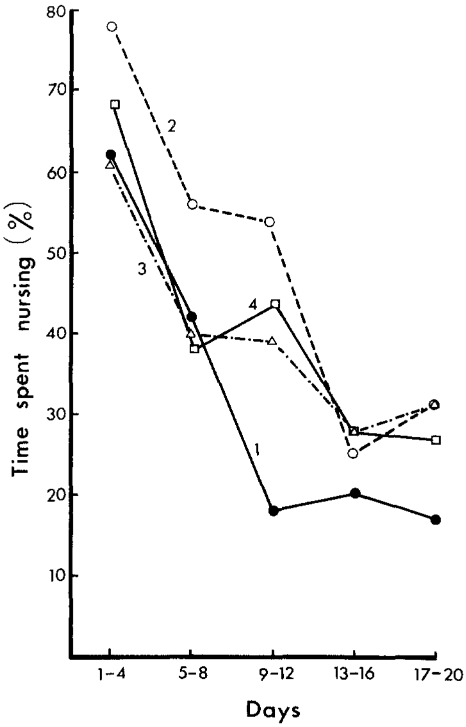

FIG. 1. Time spent nursing, as a percentage of total observation time, for consecutive blocks of 4 days from the birth of the pups. Four litters are shown; litters one and two had one and two pups, respectively; litters three and four had three pups each.

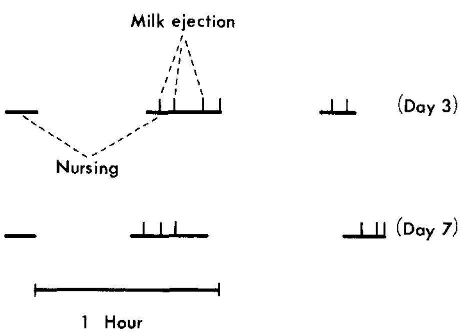

FIG. 2. Sequence of nursing and milk ejection for the mother suckling three pups, on the afternoons 3 and 7 days after parturition. On each occasion, all three pups were suckled. For all examples of milk ejection recorded, the mean interval between milk ejection was 5 min 15 sec (SD 2 min 2 sec); the mean interval from the onset of nursing to the first milk ejection was 4 min 30 sec (SD 40 sec).

The precocial birth of the young in *Acomys cahirinus* involves several special adaptations. Pregnancy is long (45 days), the litter size is small, and there are only four nipples. Labor lasts a long time, and obstetrical help is frequently given by females other than the mother (Dieterien, 1962). In contrast, the feeding of the young does not seem to involve any special adaptations, at least in gross features. The nursing posture is exactly like that found in rats and mice, and the alternation of nursing and nonnursing episodes is quite similar. The functioning of the milk ejection reflex, with milk ejections occurring repeatedly with intervals of several minutes between each is also typical of these species. Two differences of detail are of interest. In rat pups leaving the nipple at milk ejection is common at 15 days of age but quite rare before 10 days of age (Drewett, unpublished); in the precocial spiny mouse this response is found reliably on the first days of life. Second, the proportion of time spent feeding the young is higher in spiny mice than in either of the two inbred strains of mice we are studying in this laboratory (C57BL/6J and SEC/1ReJ); the young of the first of these strain are more precocial than those of the second, and the mother spends more time feeding the young in the first strain (D'Udine & Gerosa, manuscript in preparation).

Acomys cahirinus seems to us of special interest for two reasons. First, it is possible to detect the milk ejection in this species very reliably just by watching the pups. This is also possible in rats and mice, but much less reliable as the pups are smaller and get hidden by the mother and the nest. Most of the work on milk ejection in these species has therefore used conscious pups suckled on anesthetized mothers. It seems likely that the spiny mouse will be of value in the study of those aspects of the milk ejection reflex that cannot be studied in anesthetized mothers, such as its control by conditioned stimuli. Second, we are carrying out further comparisons between *Acomys cahirinus* and inbred strains of mice in the belief that it may be of interest to compare precocial murids produced by artificial and by natural selection.

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